

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (EE) Examination

November 2012

EE301 Electrical Machines-II

Date: 26.11.2012, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Multiple Choice Questions

[07]

- (i) The relation among synchronous speed (N_s), rotor speed (N) and slip (S) is _____
(a) $N=(S-1)N_s$ (b) $N=(1-S)N_s$ (c) $N=(1+S)N_s$ (d) $N=SN_s$
- (ii) The rotor winding of a 3-phase wound rotor is generally _____ connected.
(a) star (b) delta (c) partially star partially delta (d) none of the above
- (iii) If the slip of a 3 phase induction motor increases, the p.f of the rotor circuit _____
(a) is increased (b) is decreased (c) remains unchanged (d) none of the above
- (iv) The conditions of induction motor at standstill resemble those of a transformer whose secondary is _____ (a) Short-circuited (b) Open-circuited
(c) supplying a variable resistive load (d) none of the above
- (v) The circle diagram of induction motor does not provide accurate results because it is _____ (a) a graphical method (b) uses approximate equivalent circuit
(c) assumes leakage reactance to be constant (d) all of the above
- (vi) If the stator voltage of an induction drops by 10 %, the torque of the motor will drop by _____ (a) 10% (b) 40% (c) 20% (d) 30%
- (vii) The linear synchronous speed of a linear induction motor (LIM) does not depend upon _____
(a) Supply frequency (b) pole pitch (c) number of poles (d) none of the above

Q - 2 (a) Draw and explain the complete torque speed curve of induction machine.

[07]

(b) Draw and explain the phasor diagram and equivalent circuit of three phase induction motor.

[07]

OR

Q - 2 (a) Explain the power stages in an induction motor and derive the torque equation.

[07]

(b) Write a note on methods of speed control of poly phase induction motor.

[07]

Q - 3 The results of no-load and blocked rotor tests on a 400 V, 36.8 kW, 50 Hz, 3 phase, 4 pole induction motor are as under: [14]

No- load test : 400 V; 30 A; 1800 W

Blocked rotor test: 110 V; 80 A; 4000 W

The motor has a star connected stator having a resistance of 0.1Ω / phase. Draw the circle diagram and calculate: (i) full load stator current (ii) full load power factor (iii) full load slip (iv) full load torque (v) full load efficiency (vi) starting torque (vii) maximum torque (viii) maximum output power

SECTION - II

Q - 4 Explain the types of induction generator, principle of operation and its application. [07]

Q - 5 (a) A 2 pole, 240 V, 50 Hz, single phase induction motor has the following constants referred to the stator: $R_1 = 2.2 \Omega$; $X_1 = 3.0 \Omega$; $R_2' = 3.8 \Omega$; $X_2' = 2.1 \Omega$; $X_m = 86 \Omega$. [08]

Find the stator current and input power when the motor is operating at a full load speed of 2820 r.p.m.

If the iron and friction losses amount to 50 W, find (i) total mechanical power developed (ii) useful shaft power (iii) shaft torque and (iv) efficiency.

(b) Explain the working, construction, operation, characteristics and application of universal motor. [06]

OR

Q - 5 (a) Explain the shaded pole induction motor [04]

(b) Explain the capacitor start capacitor run type of single phase induction motor [04]

(c) In a double cage induction motor, if the outer cage has an equivalent impedance at standstill of $(2+j2) \text{ ohm}$ and the inner cage an equivalent impedance of $(0.5 + j5) \text{ ohm}$, determine the slip at which the two cages develop equal torques. [06]

Q - 6 Write a technical note on the following: (Attempt any TWO) [14]

(i) Schrage motor

(ii) Stepper motor

(iii) Repulsion and Hysteresis motor

(iv) Linear induction motor (LIM)
